

J1. Involvement of stakeholders and field experts

- Email Transcripts

- Initial email for Stakeholder validation

Dear Dr. Brossman,

I hope this email finds you well.

As we progress in the design process for the MIP device, we have reached a stage where stakeholder validation is essential before moving forward. Below, we have outlined the high- and low-priority requirements we have identified for the device:

High Priority Requirements

Durability/Longevity

- You previously mentioned how you would like this to last for the end of your tenure, but we are aiming to have it last for at least five years. Durability is very important, the MIP device will be passed around with a lot of people, and they are very expensive to purchase. Having a durable MIP device will allow for a long term financially-sound device that won't need to be fixed.

Accuracy

- The device must measure MIP with an accuracy of at least ± 1 cm H₂O, adhering to established medical protocols for respiratory pressure assessment. Accuracy is essential to ensure that the device provides reliable results for student training. This requirement is informed by the precision of previous design attempts.

Portability

- Portability is likely going to be the most important factor in the success of our design. The device will have to be passed around by 90+ students all using it and if it's heavy or cannot be handed off to another student easily it will not be an effective design. Being able to portably transport the device into a storage container so that the device will not move around or break.

Simlicity and Redicability

- We are going to need to make 10-20 IMT devices AT MINIMUM. If we cannot produce that many copies of the device by the end of the semester, then the device is not sufficient and must be redesigned in a simpler way. This device must also be simple enough for a college level student to understand.

Affordability

- Another important factor is the affordability of the device. The device must be able to be produced within our budget as a high school engineering class. The components of the device must be able to be acquired cheaply and in quantity.

Low Priority Requirements

Aesthetics

- The college students might not be willing to use a device if it looks highly unprofessional and cheap. Our device should be professional enough that it doesn't scare away the college students.

Digital Display

- While we would like to include a digital display, due to time constraints, we feel that we won't be able to reach the end goal. Also, with a limited budget, and experience we would not be able to achieve this. Although, the digital display would make this device very user friendly.

We would greatly appreciate your feedback on these priorities. Apologies for the lengthy email—I wanted to provide sufficient detail for clarity. I've also [attached the full document if you would like to review](#). If there's anything else you need or any additional input you'd like to provide, please let us know!

Best Regards,
Sachita Upadhyia (Ryan Schwarz, Justin Bischoff, Hunter Discala, and Lily Goldberg)

- Response:

 **Brossman, Heather**  hb456@drexel.edu
to me 

Hello Team,

This is well done. I would offer that portability is not a huge issue as we can have an identified place for this device to sit (on a counter depth counter or on a mobility cart) if it is a larger device. The students can "move to it", it would have to be able to be portable in the sense that it can get from your HS to Drexel University (fit in an elevator, car trunk ect.). I am still working on a budget- but I don't expect a large dollar amount. This I will keep working on!

Dr. Brossman

Heather Brossman PT, DPT,DHSc
Board certified in pediatrics and cardiovascular and pulmonary physical therapy
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*Please forgive the typos as this is send from my phone *

On Nov 19, 2024, at 11:37 PM, Sachita Upadhyya <upadhyya.sachita@gmail.com> wrote:

EXTERNAL.

- Follow up email and her response:

 **Sachita Upadhyaya** <upadhyaya.sachita@gmail.com>
to Heather ▾
Dec 17, 2024, 12:02 PM ☆ ☺ ↶ ⋮

Hello Dr. Brossman,
I hope this email finds you well.
Thank you so much for your feedback. Before we proceed with our prototype, we wanted to ask you about your preference regarding the units of measure for the barometer. I know that your existing devices use centimeters of water per inch, but if there is anything else you would prefer, please let us know.

Thank you,
Sachita (and team)

xxx

 **Brossman, Heather** <hb456@drexel.edu>
to me ▾
Dec 18, 2024, 8:52 AM ☆ ☺ ↶ ⋮

Hello Sachita,
If we can be between 2-3 that is great as there is no need to be more precise.
Dr. Brossman

Heather L. Brossman PT, DPT, MS, DHS, (She/Her)
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 **Drexel**
UNIVERSITY

- Feedback

- October 29, 2024

- What her needs are, the current problems, and what the MIP is used for.

- Weight training for the respiratory muscles analogy - how to know when the limits to achieve the purpose of exercise (ex: strengthening muscles)
- Diaphragm
 - Has a dome shape, helping contract and expand the lungs
 - Change of pressure helps the air go from the atmospheric pressure to bodily pressure
 - Two muscles together make it
- To help ventilate:
 - Make diaphragm stronger
- Weaken the organ:
 - Injuries from surgeries
 - Car crashed
 - Kicked
 - Obesity
 - Cachexia
 - Neuromuscular disease like ALS, multiple sclerosis
 - COVID-19
- MIP

- Brings the air in during therapy
- Tears the tube by taking everything out
- Has the 1500 power device, but that might be another problem (she needs more of them)
 - Hard to clean
- MAP
 - Bring the air out
- Strengthening Inspiratory Muscles
 - What we have
 - Super expensive
- Want the scale to reach 200 (max pressure)
- **What do we, as PWHS student, need to do?**
 - Need to teach respiratory measurement for clinicians
 - Need reasonable cost measurement device
 - Need ease of use for cleaning
 - A lack of accessible devices
- **Our Questions**
 - Would it be easier to have cleaning stations (15 devices per station) or easier to have detachable mouth pieces?
 - Combination of the two, but she really wants our opinion on it since it is a lot of pieces.
 - How precise does it need to be (to the nth degree)?
 - She not doing research about it, so she wants to know that it is reliable enough so that, with multiple trials with similar-ish numbers the outcomes are the same
 - Precision over accuracy
 - Does the needle need to stay where it is so that people can read? Screen better?
 - Definitely has to stay because it is very hard to miss. Stay around the peak number.
 - How portable?
 - Doesn't have to be super portable
 - How many years is this going to be used for?
 - Love this to last for 15 years
 - Shouldn't have too many drops, but make sure that it is durable enough
 - Some level of protection

- What is the intended user experience for students? Should the device be easy to use independently, or will there be assistance from instructors during measurement?
 - Individually make it more interesting for everyone
 - Cleaning it is becoming a problem since there is a large number of people moving around
- What are the specific sanitation protocols?
 - Nothing yet
- Are they giving us the funding to build this?
 - Pending
- Is the goal to improve an existing MIP device or create a completely new one?
 - A HUGE Improvement or create a new one
- Are there specific materials wanted to ensure easy and deep cleaning?
 - Something that can withstand a limited amount of water
- Will it work with software for data analysis also what formats will it support?
 - Up to us
- Will the device be digital or analog, and what are the pros and cons of the chosen format?
 - Up to us
- **Field Expert**
 - **Calibration:**
 - The experts told us that in terms of calibration, we should aim for accuracy, but it doesn't need to be perfect. They wanted us to make sure that, when checking the calibration, we use a known reference point/device and it should be as close as possible to the real pressure.
 - **Durability:**
 - We want this to be able to last about 5 years, so both experts highlighted the importance of minimizing the number of moving or fragile parts to improve durability.
 - **Learning Objectives:**
 - The experts emphasized that the goal of this project is not only to provide accurate measurements but also to help students learn from the process. They suggested that if a

digital display is included, it should serve as a tool to help students interpret the data easily.

J2. Comprehensive Evaluation

Make sure that you provide clear questions/prompts that will elicit feedback from your stakeholders and field experts on several aspects of your design and several parts of your engineering design process.

Questions For Dr. Brossman:

- What do you think of our device?
- Do you think it will be able to satisfy your needs?
- What are some things you don't like about it/what can we improve?
- How would you be willing to pay per device made?
- Would you prefer our old design or the new design?